



**Department of Electrical, Computer
& Biomedical Engineering**
Faculty of Engineering & Architectural Science

Course Title:	Low Power Digital Integrated Circuits
Course Number:	ELE734
Semester/Year (e.g. F2016)	F2025

Instructor:	Dr. Andy Gean Ye
Lab Report Number	Lab #1 - Post Lab
Lab Title	Lab 1 - PostLab: Characteristics of MOSFET Devices

Due Date	October 15, 2025
Submission Date	October 15, 2025

Last Name	First Name	Student Number	Section	Signature*
Bhuiyan	Hasib	*****9402	05	

*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab report is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: <http://www.ryerson.ca/senate/current/pol60.pdf>

Introduction:

The lab assignment for Lab 1 introduces the concept of characteristics and attributes of MOSFET Devices. The lab looks into NMOS and PMOS transistors, specifically how to create and initialize them in Cadence Virtuoso; this includes how to alter the specifications of the transistors, labelling and assigning pins, how to create a mistake-proof schematic, and to create different views of the cell. The lab heavily emphasizes on creating different views of the NMOS cell, including schematic, layout, symbol and calibre at the end. With the use of all these cell views, we create a standard transistor model with schematics, layout, design rule verifications, LVS check, and PVE(Parasitic Extraction), comparing the schematic to the actual layout and verifying that all netlists and parameters are functional, ultimately producing a professional mistake free model. Using testbenches and circuits we also look into the I-V characteristics of NMOS and PMOS, intrinsic device capacitances and second order effects. Using the Cadence Virtuoso ADE L, we are able to produce multiple graphs for analysis, comparing different voltage and current values, to understand concepts like channel length modulation, I-V behaviour, body effect, and subthreshold conduction. This lab poses as an introduction to the tools and use of Cadence Virtuoso.

Post-Lab Results:

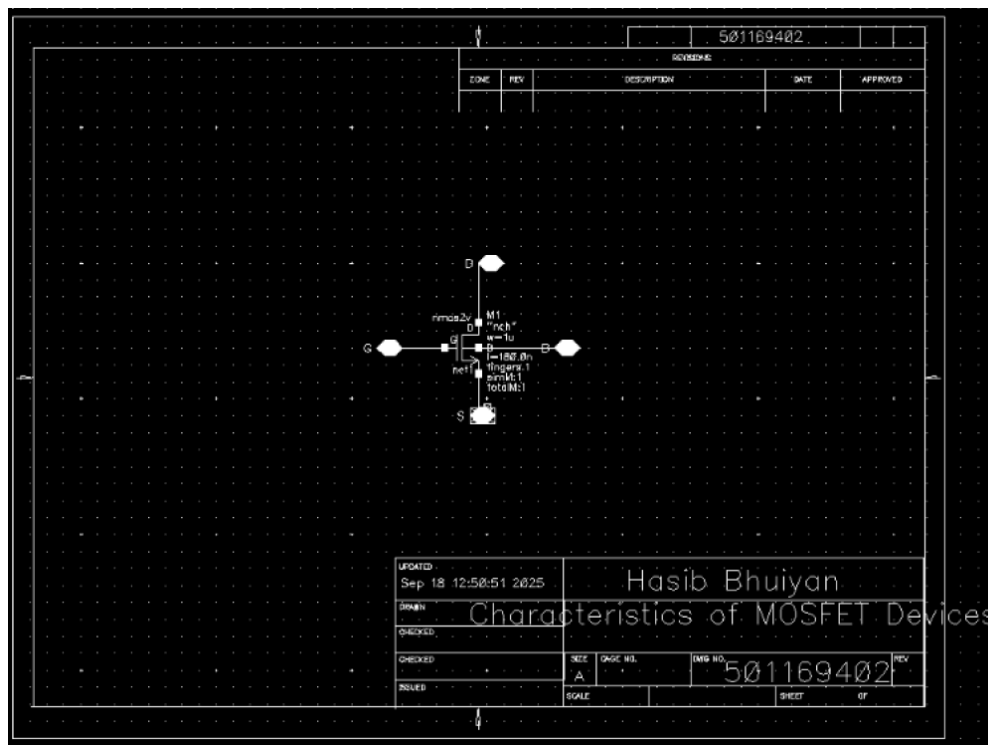


Figure 1: Schematic representation of NMOS

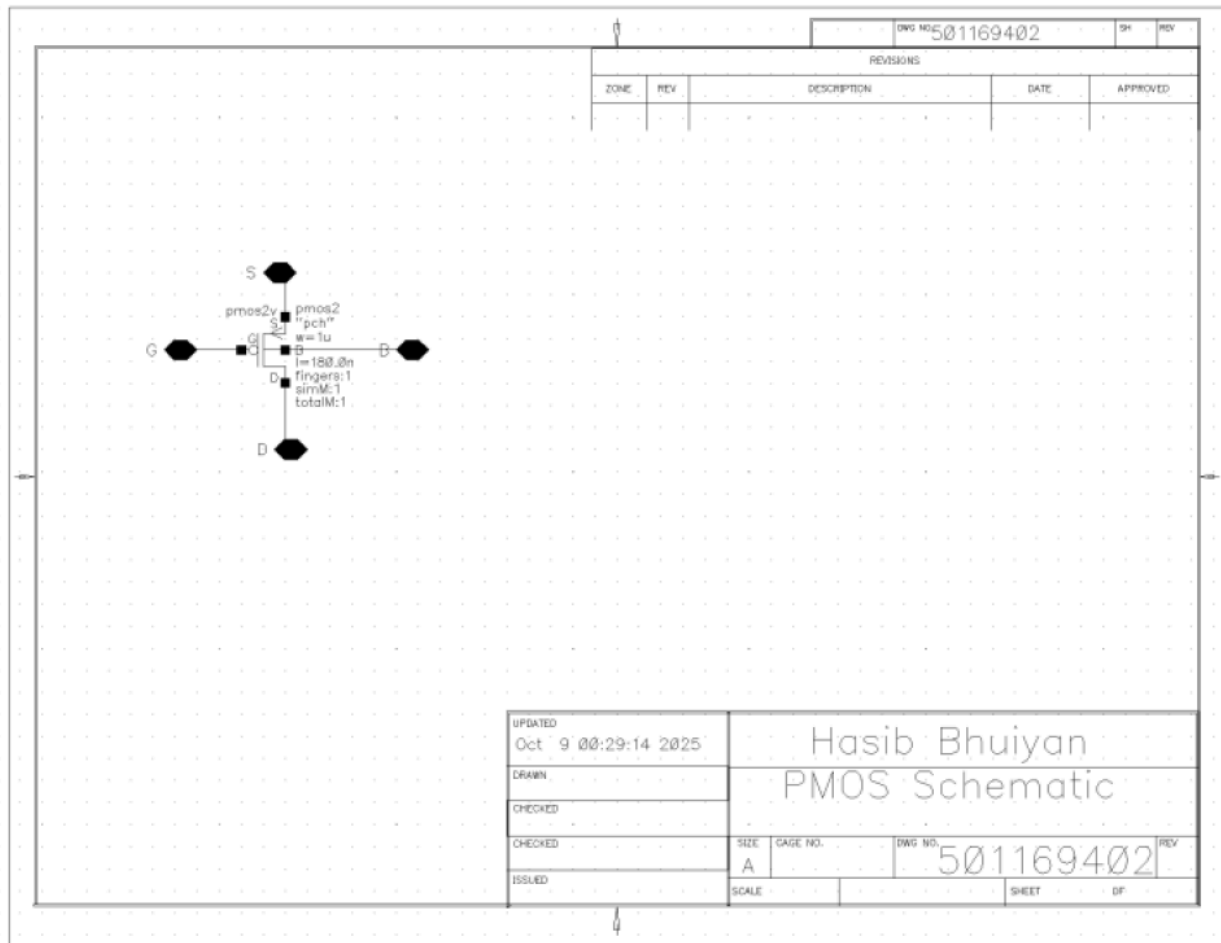


Figure 2: Schematic representation of PMOS

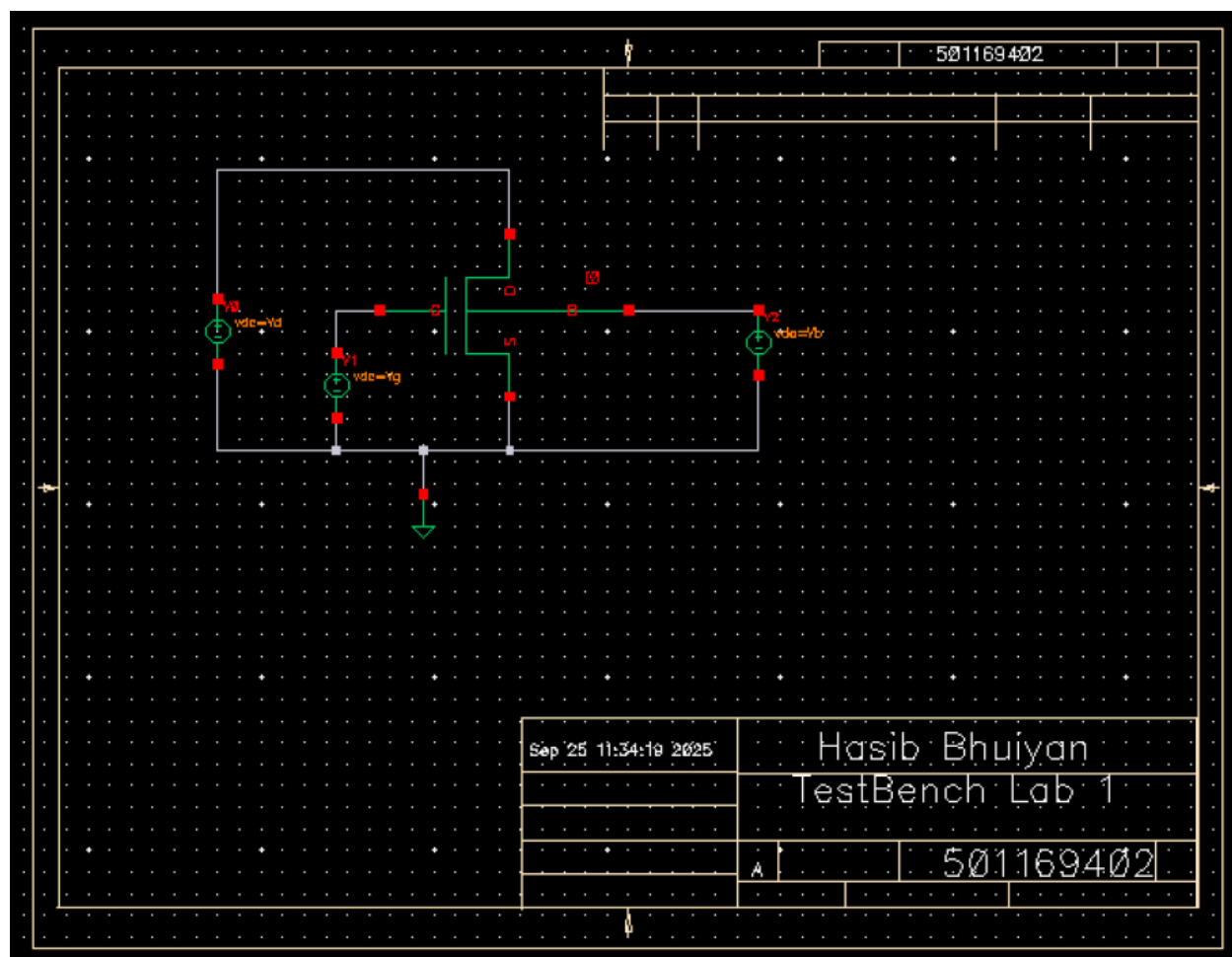


Figure 3: Circuit Schematic for NMOS Testbench Cell

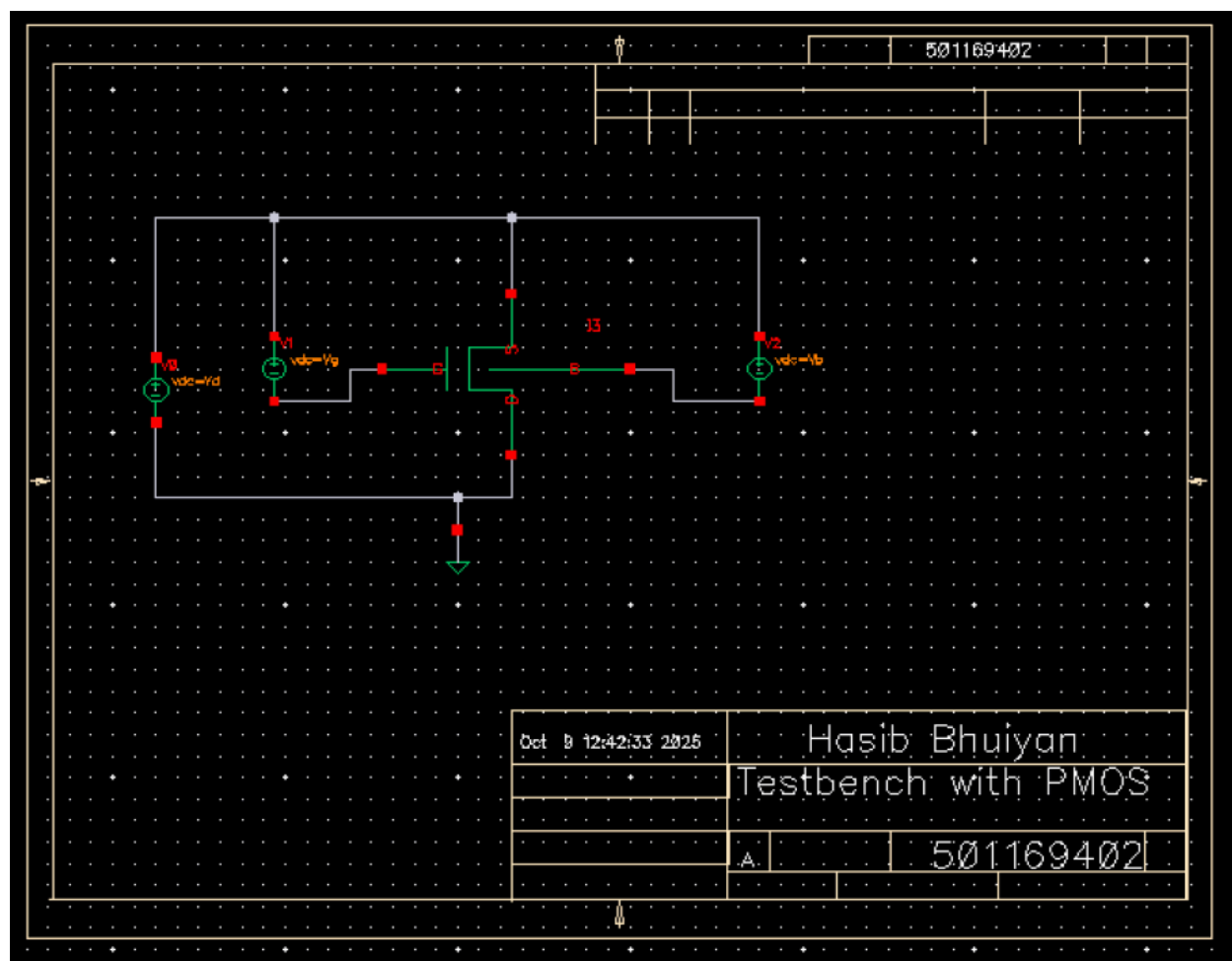


Figure 4: Circuit Schematic for PMOS Testbench Cell

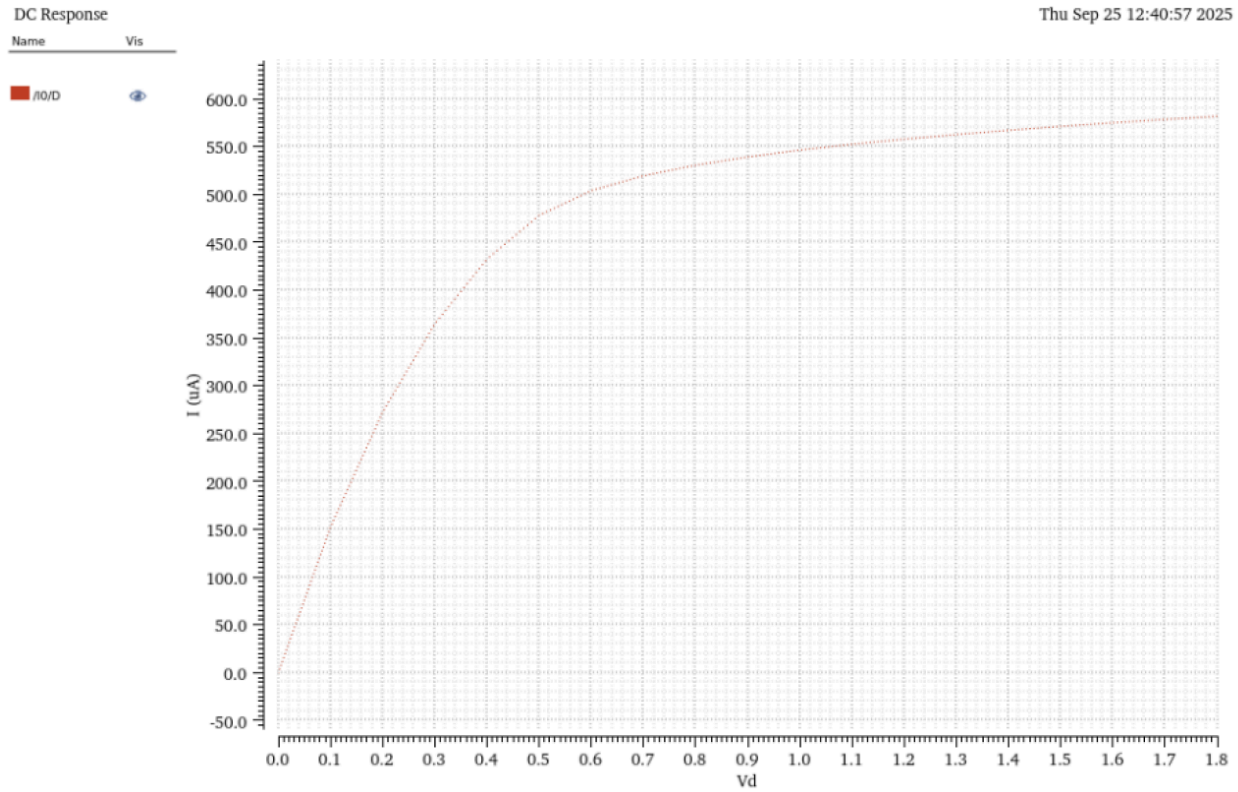


Figure 5: I-V characteristics of NMOS(Drain Current when $V_{gs} = 1.8\text{V}$)

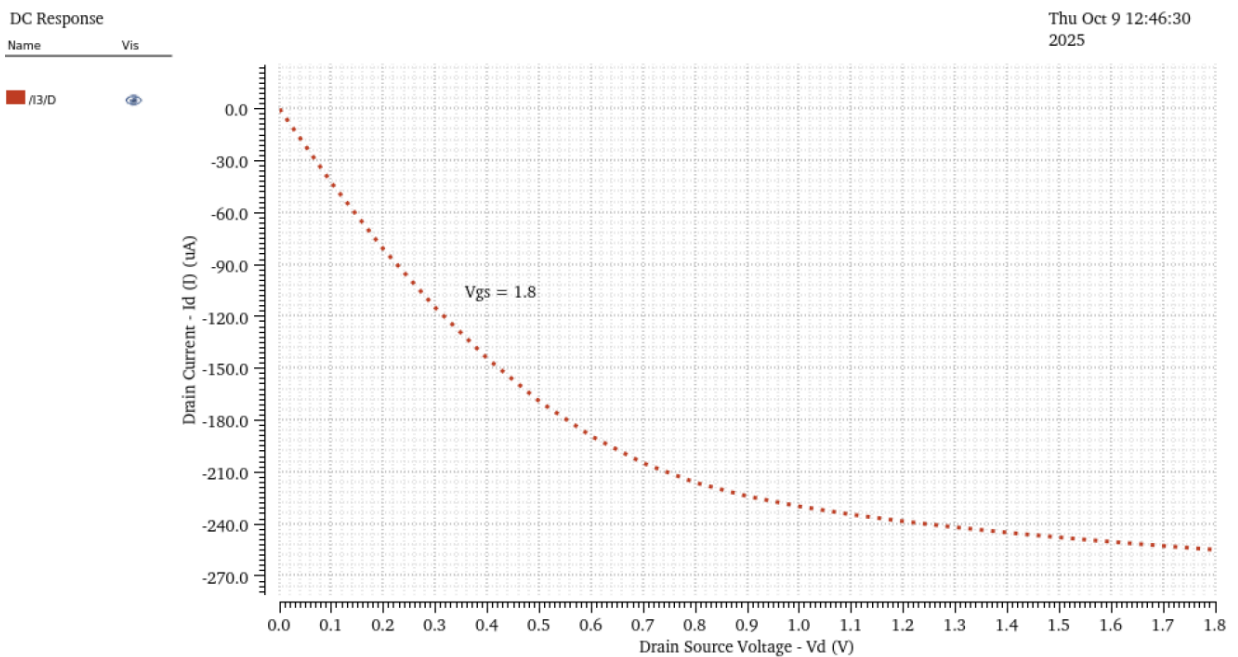


Figure 6: I-V characteristics of PMOS(Drain Current when $V_{gs} = 1.8\text{V}$)

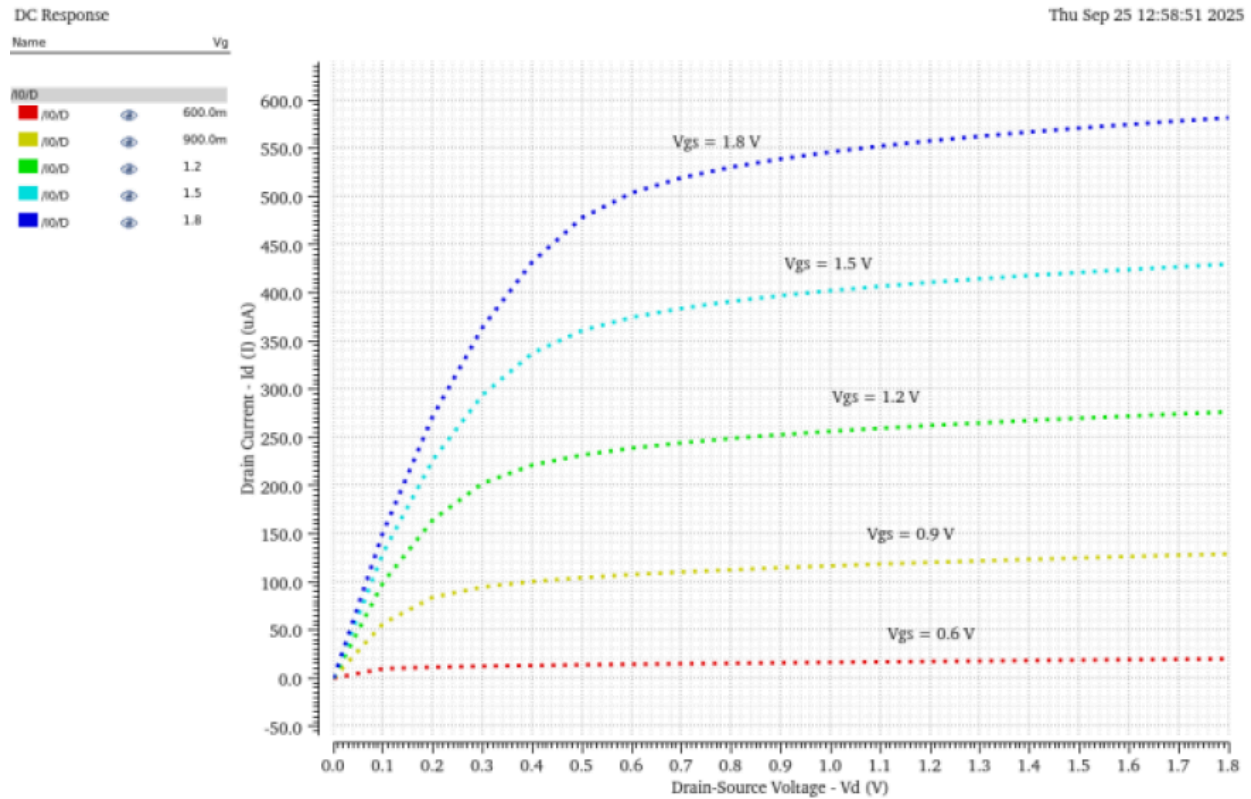


Figure 7: I-V characteristics family of NMOS(multiple values of Vgs)

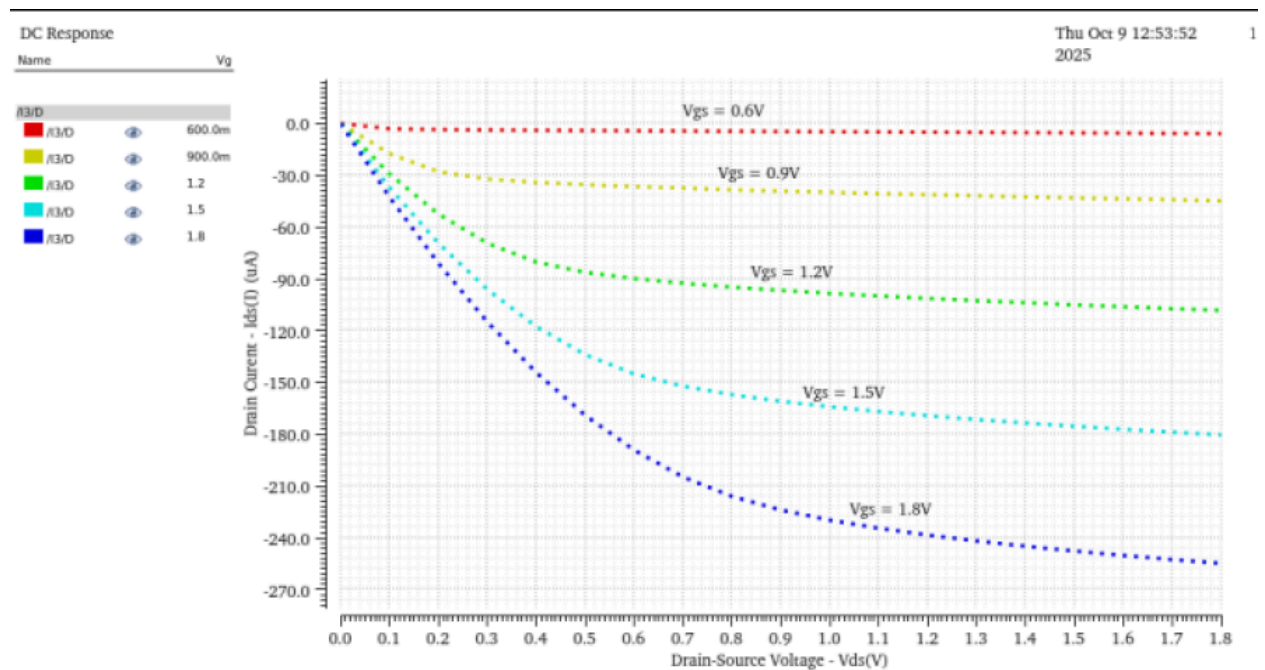


Figure 8: I-V characteristics family of PMOS(multiple values of Vgs)

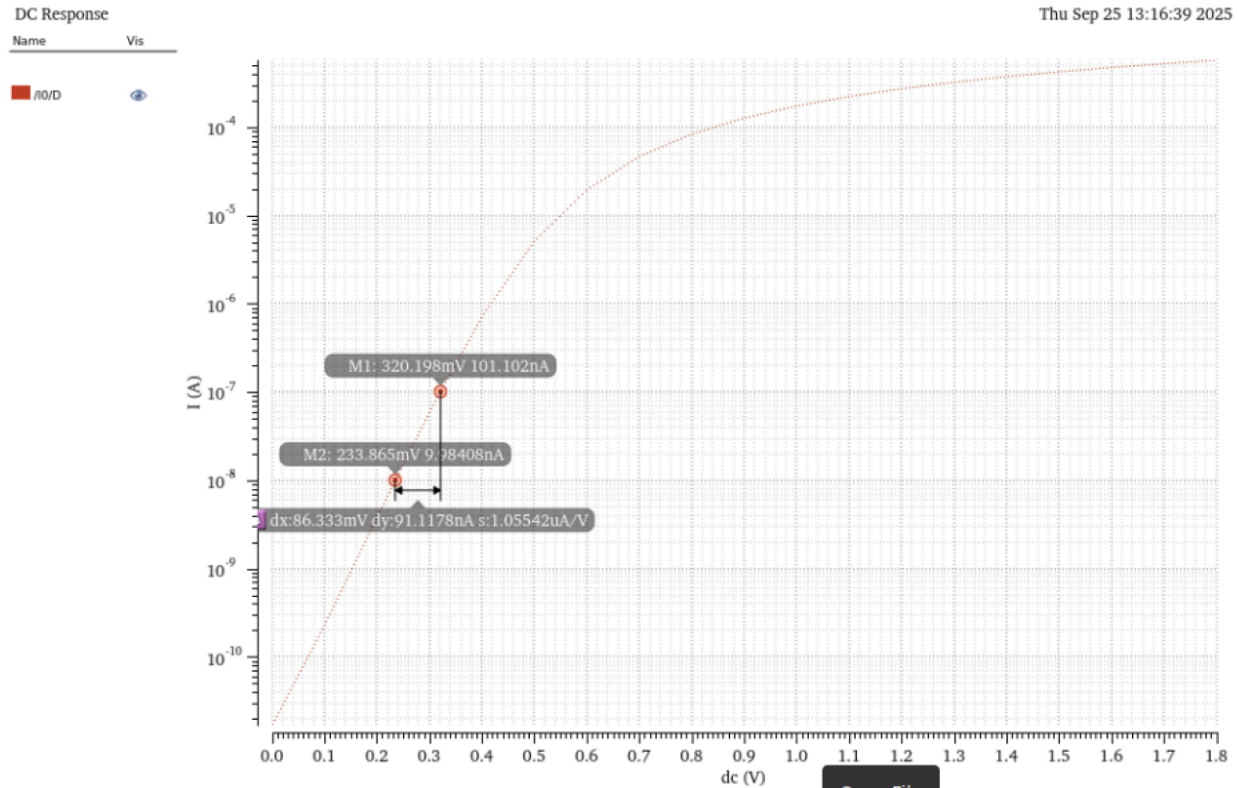


Figure 9: I_{ds} vs V_{gs} graph curve of NMOS with analysis of subthreshold conduction and slope

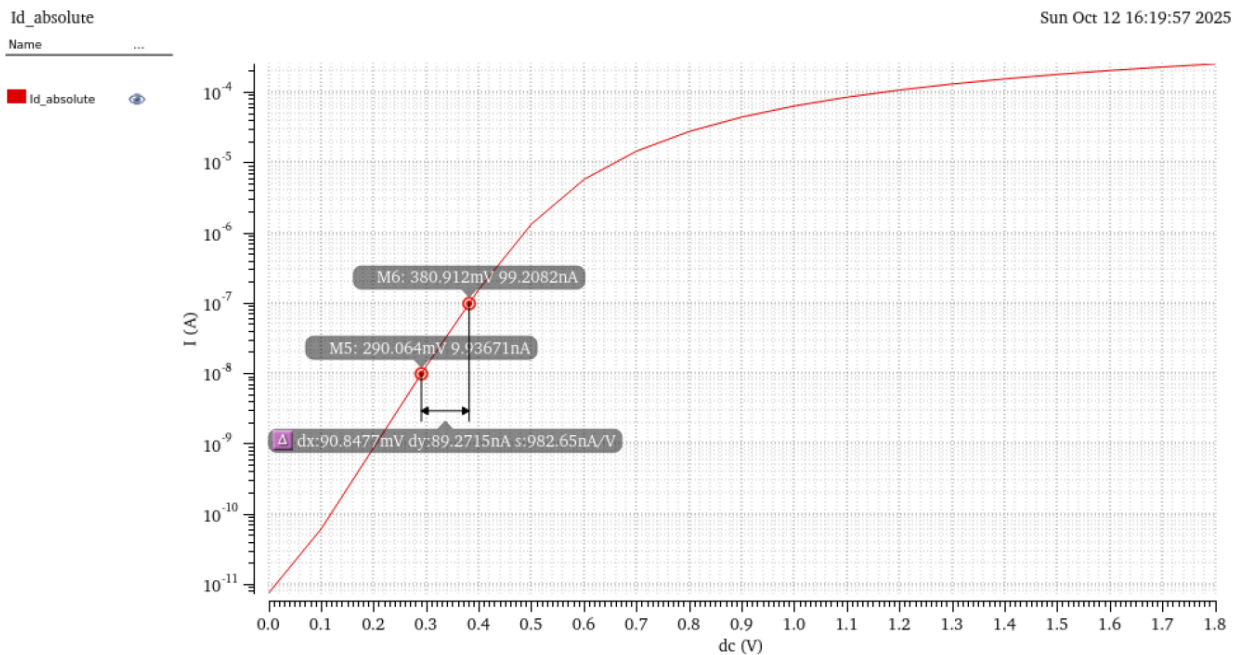


Figure 10: I_{ds} vs V_{gs} graph curve of PMOS with analysis of subthreshold conduction and slope

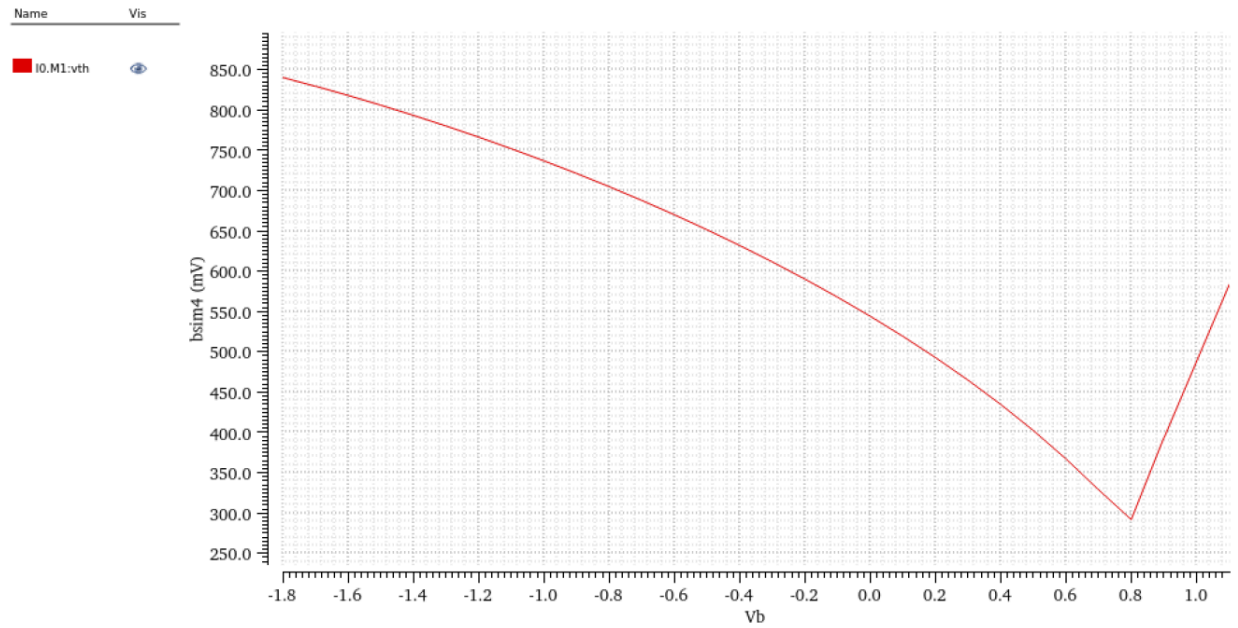


Figure 11: V_{th} vs V_b graph curve of NMOS for analysis of body effect

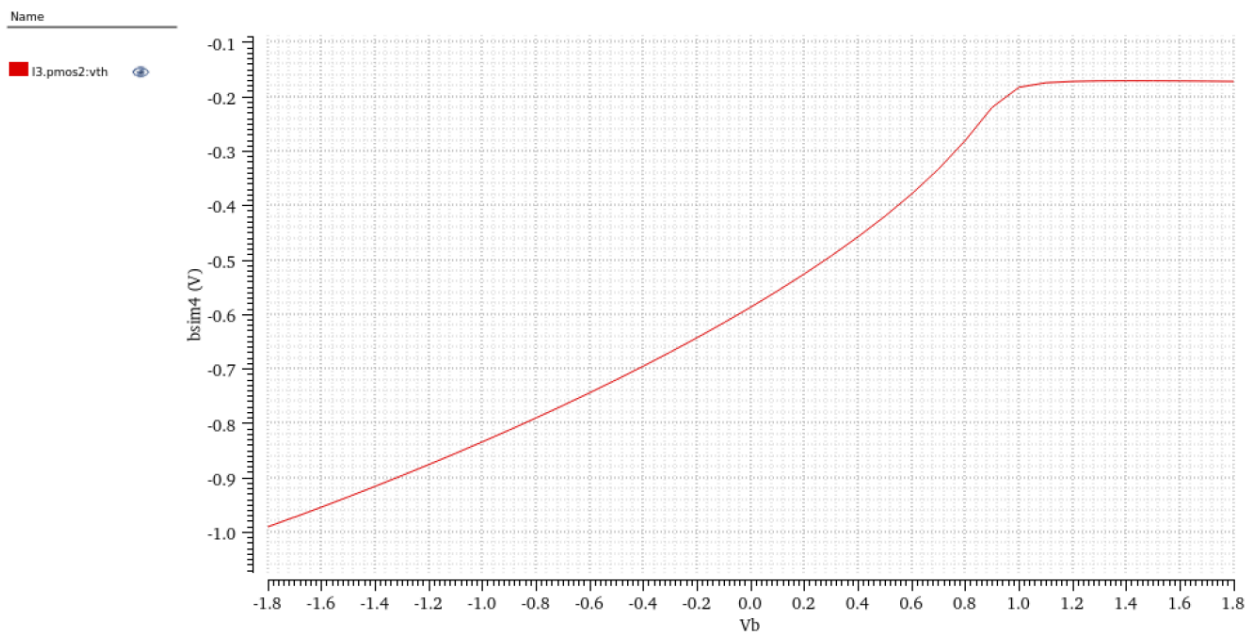


Figure 12: V_{th} vs V_b graph curve of PMOS for analysis of body effect

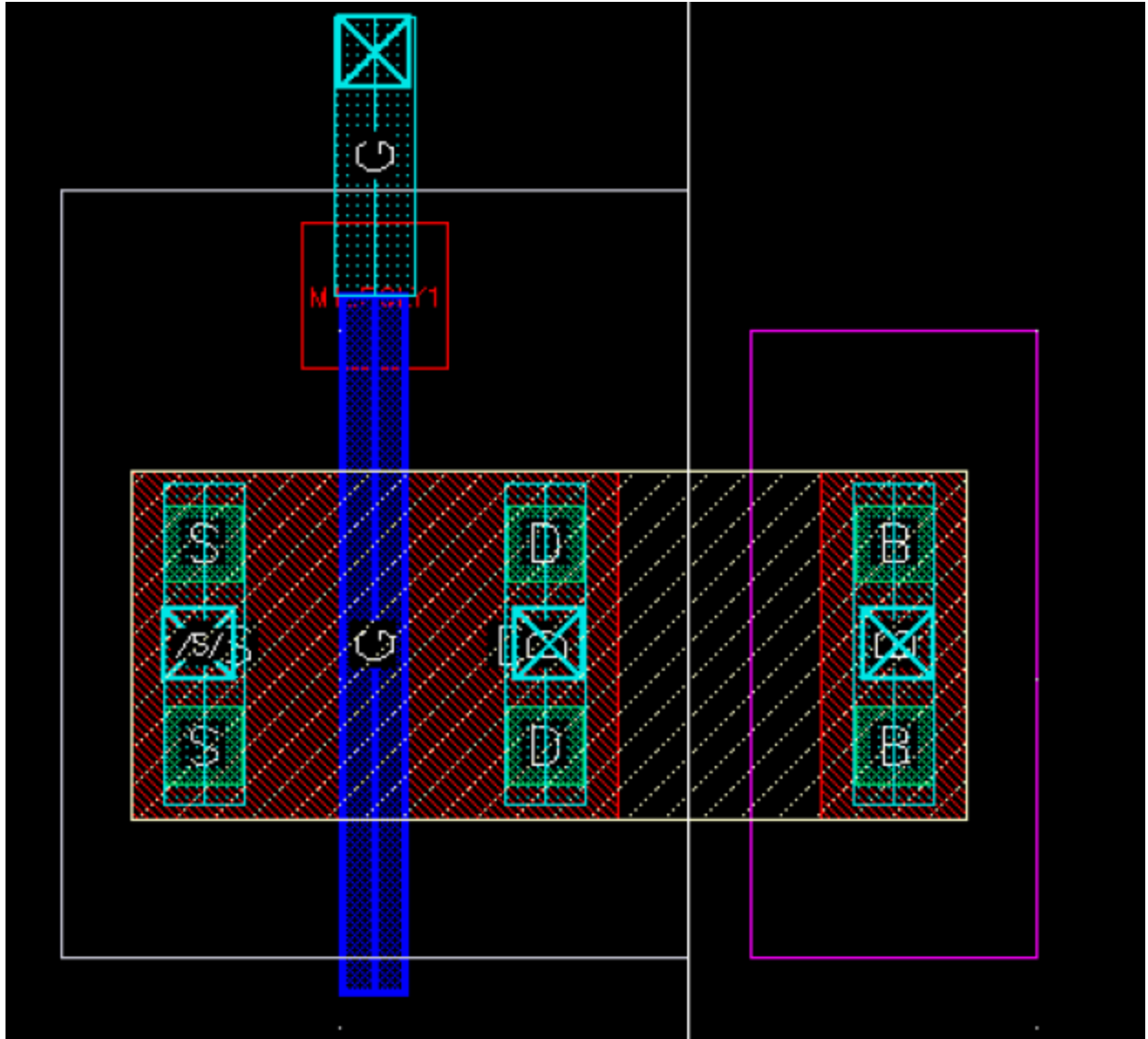


Figure 13: Full Layout view for NMOS

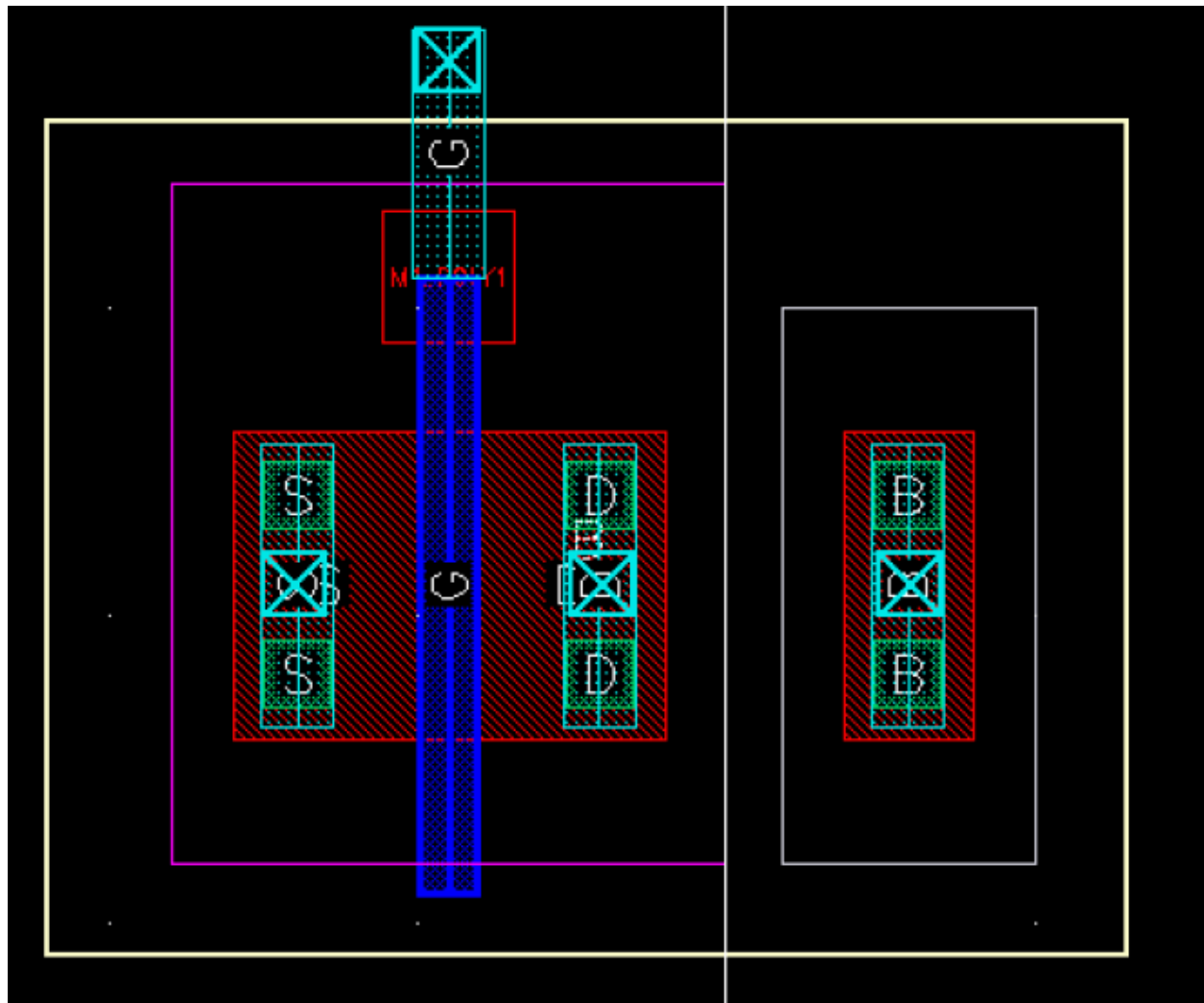


Figure 14: Full Layout view for PMOS

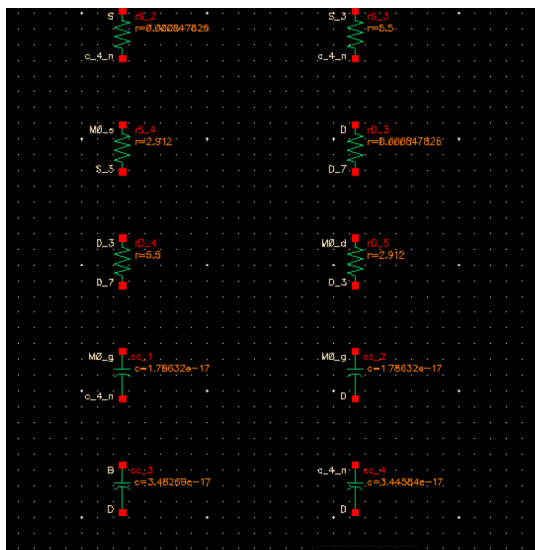
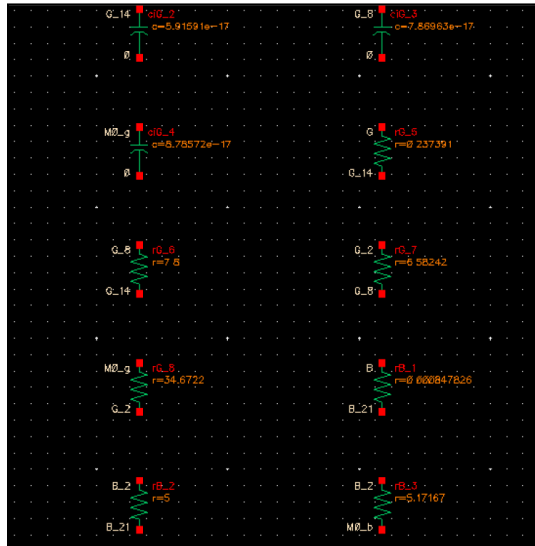


Figure 15: PVE(Parasitic Extraction) Extracted view of schematic for NMOS transistor

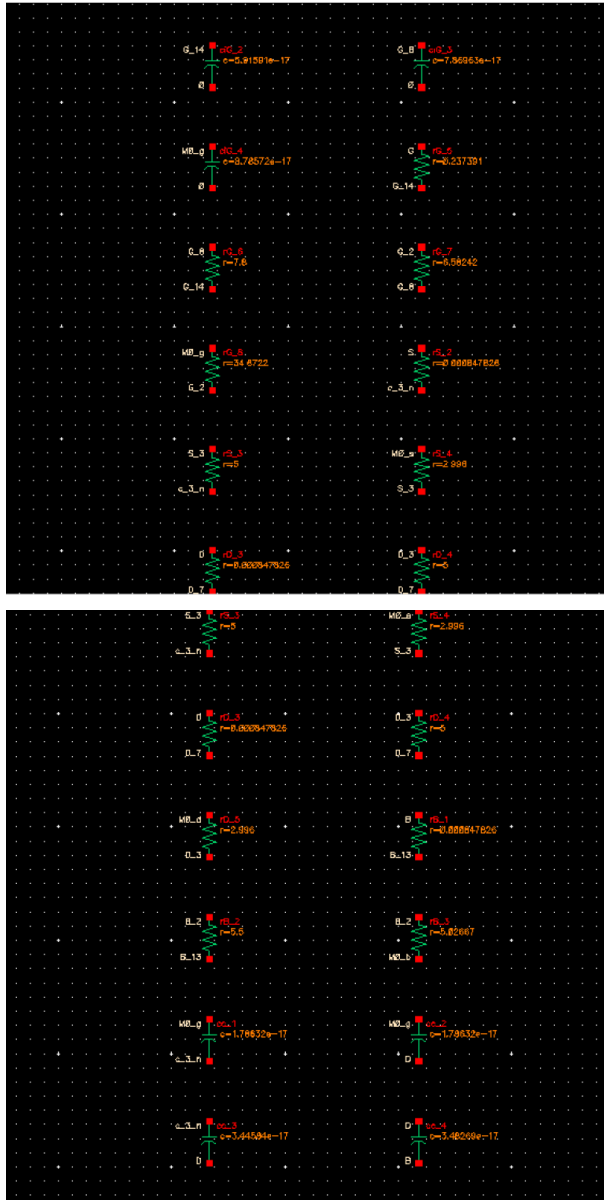
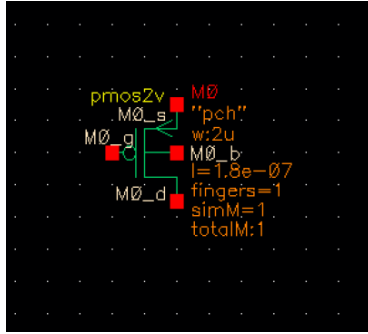


Figure 16: PVE(Parasitic Extraction) Extracted view of schematic for PMOS transistor

In the lab, different characteristics and behaviours of both the NMOS and PMOS transistors were analyzed using the graph simulations and measurements. Using these simulations and measurements from I-V characteristics, subthreshold behaviour and slope(I_{DS} - V_{GS} graphs), and body effect (V_{TH} - V_B graphs) we were not just able to analyze the characteristic of the transistors, but we were also able to determine differences and variations from an ideal case and a realistic case of transistor behaviour. By comparing the original schematic results with the extracted calibre post-layout results using PVE(Parasitic Extraction), we were able to find the effects and conditions that must be taken into consideration in a practical real transistor.[Do note that post-layout simulations were not successful due to cell naming issues; calibre file was assigned to 'NMOS' cell, while the layouts, schematics and symbol belonged to 'nmos' cell, but other than that the pre-layout simulations, DRC check, LVS check, and PVE execution are fine]. We could see that the behaviour of the graph and type of curvature of the post-layout simulation is not very much different and very consistent with the pre-lab theoretical behaviour and graph curve, proving that both the theory and the post-layout results are both appropriate behaviours. From the I-V characteristics of the schematic(pre-layout), we could see that it follows through with the theory, as there is a linear increase in I_d (drain current) with V_{ds} (drain voltage) in the triode region just before $V_{ds} = V_{gs} - V_{th}$, and after V_{ds} reaches that point, it goes through the saturation region beyond it. The I_{DS} - V_{GS} graph was used to calculate subthreshold behavior and slope; it showed exponential increase when $V_{gs} < V_{th}$ till it reached V_{th} proving the existence of subthreshold conduction, minor current increasing and running even when V_{gs} doesn't reach V_{th} , this complies with the theory in pre-lab subthreshold conduction model. The V_{TH} - V_{SB} plots prove the concept of body effect; that threshold voltage increases with larger source-to-body bias, agreeing with the theoretical equation in pre-lab that relates V_{th} with V_b . Now the difference between the post-layout and pre-layout is in the accuracy of the simulation, as the post-layout simulation takes external factors into play that change the curve and the numerical value slightly. Factors like additional resistances, capacitances within interconnections and device layout positions and geometry, that are not accounted for in an ideal pre-layout scenario. For the I-V characteristics, it could be seen that the triode regions and saturation regions are slightly reduced and deviated from the ideal behaviour; minor discrepancies occur with concepts like channel length modulation(CLM), which creates a slope as opposed to the perfect horizontal current saturation line. Note that CLM and other parasitic factors play a bigger impact when dealing with short-channelled devices, showing bigger deviations. Now although the difference between the post-layout and pre-layout simulation results is minimal and there are no changes in the actual behaviour and curvature, the slight difference can still be considered significant as transistors work very precisely and small parasitic factors can still make a recognizable and notable impact.